

CSA15 - CLOUD COMPUTING AND BIG DATA ANALYTICS

INDIVIDUAL CONTRIBUTION AND REFLECTION

Field	Student Entry
Slot	B
Assignment Title	Cloud-Ready Smart Campus Service Platform: API Implementation, Workload Sizing and Virtualized Deployment
Team Number	3
Date of Submission	02.09.2026
GitHub Repository	https://github.com/192421053simats-cmd/Cloud-ready-smart-campus-service.git

TYPE DIRECTLY IN THIS DOCUMENT. Replace all bracketed placeholders with original content. Do not reproduce the assignment question, detailed instructions or another team's text.

Field	Student Entry
Student Name	Blessy S
Register Number	192421053
Team Number	3
Assigned Module / Responsibility	API implementation and workload sizing
Team GitHub Repository	https://github.com/192421053simats-cmd/Cloud-ready-smart-campus-service.git

1. Work Personally Completed

Personal Contribution
I worked on the backend API implementation and workload sizing for the Cloud-Ready Smart Campus Service Platform. My contribution included defining the REST API flow for authentication, resource search, booking creation and booking retrieval; preparing authentication and role-based authorization middleware; validating booking input; and handling appropriate HTTP status responses. I also worked on the workload-sizing calculation using the estimated campus users and peak-hour request pattern. The sizing resulted in an estimated peak traffic of about 3.5 requests/second and supported the selection of a 2-vCPU, 2-GB RAM and 20-GB disk virtual machine. I also contributed to the Ubuntu Server LTS and optional Docker deployment decision.

Evidence Type	File / Module	Commit / URL	What It Proves
Code	backend REST API / API routes	https://github.com/192421053simats-cmd/Cloud-ready-smart-campus-service.git	Shows the API implementation for authentication, resource access, booking creation and booking retrieval.
Design / Sizing	Workload sizing and deployment decision	https://github.com/192421053simats-cmd/Cloud-ready-smart-campus-service.git	Shows the peak workload estimate of approximately 3.5 req/s and the selected 2-vCPU, 2-GB RAM, 20-GB disk VM.
Test	API and deployment evidence	https://github.com/192421053simats-cmd/Cloud-ready-smart-campus-service.git	Supports API testing and VM/deployment evidence; actual screenshots should be attached from the team's implementation.

2. Individual Implementation and Result Evidence

Implementation Evidence
Implementation evidence from the assignment includes the REST API routes and authentication/authorization logic, workload-sizing calculation, and VM deployment configuration. Insert the actual screenshots/outputs from the team's GitHub, API testing tool and VM here before submission; no screenshots were available in the source assignment.

Test Performed	Expected	Observed	Correction / Decision
Authentication / token test	A valid token is returned for valid credentials.	Token returned successfully.	Kept token-based authentication as the API access control mechanism.
Resource search API	GET /api/resources returns a valid resource	Resource list returned with a valid JSON	Used authenticated and role-authorized access

	list.	response.	for resource retrieval.
Booking creation API	A valid resource/time request creates a booking and returns a booking ID.	Booking created and booking ID returned.	Used input validation and HTTP 201 for successful creation.

3. Technical Decision

Decision and Evidence
I selected an Ubuntu Server LTS virtual machine as the deployment environment, with Docker as an optional packaging layer. The alternatives were a more complex container/cloud deployment or a less controlled local-only setup. The VM was preferred because the assignment required clear CPU, RAM and disk allocation and straightforward virtualization evidence. The workload calculation estimated approximately 3.5 peak requests/second, leading to a practical prototype configuration of 2 vCPUs, 2 GB RAM and 20 GB disk with headroom.

4. Reflection

Learning Achieved
Through this contribution, I learned how a REST API can be organized around authentication, authorization, resource access and booking operations. I also learned that infrastructure selection should be based on an estimated workload rather than choosing resources without justification. The workload calculation helped me connect user activity, peak-hour requests and VM CPU/RAM/storage requirements. I also developed engineering judgement about using a virtual machine for controlled deployment and keeping secrets outside source code.

Improvement Proposed
The main improvement I would propose is to replace the estimated workload assumptions with measured load-test results before production deployment. This matters because the current peak request rate is an estimate. I would validate the improvement by running a controlled load test, monitoring CPU, RAM and API latency, comparing the measured request rate with the estimated 3.5 req/s, and then resizing or scaling the deployment if the measured demand requires it. HTTPS, stronger secret management and managed database/storage services should also be added for production.

5. Individual Declaration

Originality and Responsibility
I confirm that the contribution, reflection and evidence described in this document represent my assigned work and understanding of the team assignment. I am responsible for verifying the API outputs, workload-sizing calculations, deployment details and repository contents before submission. AI assistance was used for drafting and organizing explanations, as acknowledged in the team assignment, and the final content must be checked against the actual implementation.

Field	Student Entry
Student Signature	Blessy S
Date	02.09.2026

SUBMISSION CHECK: Typed Word document; personally written content; module evidence; working GitHub/commit links; similarity at or below 20%.